RE-MESH:
A SELF-HEALING MESH NETWORK ARCHITECTURE FOR RESILIENT POST-DISASTER URBAN COMMUNICATION

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Project Proposal Report

B.Sc. (Hons) Degree in Information Technology Specializing in

Computer System and Network Engineering

Department of Information Technology

Sri Lanka

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DECLARATION

I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

Name	Student ID	Signature
LAKSHITHA D. H.	IT22617446	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

	Name	Signature	Date
supervisor	Mr. Uditha Dharmakeerthi		29/08/25
co-supervisor	Mr. Amila Senarathne		29/08/2025

Abstract

Reliable communication is a critical requirement during disaster situations; however, conventional infrastructures such as cellular networks, satellite links and internet backbones are often disrupted or rendered completely unavailable. This project proposes the design and implementation of a resilient disaster recovery communication system using ESP32-based portable devices interconnected through Wi-Fi mesh networking. Each device operates on rechargeable battery power and supports text communication, voice transmission and centralized monitoring, enabling emergency responders and communities to maintain situational awareness even in the absence of traditional communication channels.

The proposed system introduces a distributed coordination model, where all devices function as both participants and potential coordinators, eliminating single points of failure. A PC-based offline dashboard provides centralized management, including real-time monitoring of active/inactive devices, configuration updates and a backup & restore mechanism that ensures seamless replacement of failed nodes with retained settings. Unique device identification is achieved through hardware-based addresses, enabling accurate tracking and logging of communication exchanges.

Compared to existing solutions such as satellite phones, professional radio networks and LoRa-based text systems, the proposed system offers a low-cost, scalable and fault-tolerant alternative with multimedia support. By integrating embedded system design, wireless networking and centralized control principles, this research contributes a novel and practical framework for disaster-resilient communication. The expected outcome is a functional prototype validated through fault-tolerance testing, demonstrating improved reliability, flexibility and accessibility in emergency communication scenarios.

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1. INTRODUCTION

1.1. Background of Research

Communication plays a vital role in coordinating disaster response and recovery. During natural disasters such as earthquakes, floods, hurricanes or tsunamis conventional communication infrastructures including cellular networks, internet backbones and power grids often fail. The lack of reliable communication channels severely limits the ability of emergency responders, volunteers and affected communities to share information, request assistance and coordinate rescue efforts.

Existing solutions such as satellite phones and professional-grade radio networks provide partial relief but remain expensive, require specialized infrastructure or are inaccessible in resource-limited regions. Furthermore, such systems often lack scalability and adaptability in rapidly evolving disaster environments. Therefore, there is a pressing need for an affordable, resilient and decentralized communication system that can be deployed quickly and operated without dependence on external infrastructure.

Recent advancements in low-cost microcontrollers and wireless mesh networking present new opportunities. The ESP32 microcontroller with integrated Wi-Fi, processing capability and support for mesh networking offers a promising platform to build portable, battery-powered communication nodes. By leveraging mesh networking devices can form self-healing infrastructure-free networks enabling communication even when traditional systems are unavailable.

1.2 Literature Survey

1.2.1. Disaster Communication Challenges

Research on disaster management consistently highlights the central role of communication in effective emergency response. Failures in communication are often attributed to three interrelated categories:

technological, **sociological** and **organizational** challenges (Manoj & Baker, 2007).

- Technological challenges include infrastructure damage, network interoperability issues and the urgent need for rapidly deployable alternatives when conventional systems fail.
- Sociological challenges involve public trust in information sources, the psychological volatility of affected populations and the need for context-sensitive, culturally appropriate communication.
- Organizational challenges stem from the difficulty of coordinating diverse agencies, sharing information across jurisdictions and synchronizing response operations.

A recurring theme in the literature is information overload. Disaster response agencies often face not a scarcity of data, but an excess of fragmented and unstructured information, complicating decision-making and slowing response times.

1.2.2. Mesh Networking in Emergency Response

Mesh networking technologies have attracted substantial attention as resilient communication alternatives during disasters. Studies underscore their self-organizing, self-healing and infrastructure-independent characteristics, which allow communication networks to continue functioning even when central infrastructure is destroyed.

Practical implementations illustrate this potential. For example:

- The Mesh SOS system developed at the University of Hawaii demonstrated significant performance improvements in delay, delivery ratio, robustness and energy efficiency for IoT-enabled emergency response.
- The Serval Project successfully deployed smartphone-based mesh networks during the 2010 Haiti earthquake and Hurricane Sandy, proving the viability of ad hoc mesh solutions in real crises.

These findings reinforce the suitability of mesh networking for rapid deployment and resilient communication under unpredictable disaster conditions.

1.2.3. ESP32-Based Communication Systems

Recent studies have explored the ESP32 microcontroller as a low-cost, power-efficient platform for disaster communication. Research from De La Salle University (Philippines) demonstrated an ESP32-based mesh messaging system capable of transmitting short messages and GPS coordinates within a 46-meter range, sustaining 16 hours of active operation.

Further experimental work confirmed that ESP32 devices can form self-healing mesh networks without internet connectivity or routers, using protocols such as ESP-NOW and ESP-WIFI-MESH. Technical benchmarks highlight:

- Extended range (up to 15 times longer than Bluetooth Low Energy).
- Power efficiency, balancing longer operational lifetimes with higher transmission demands.
- Automatic reconfiguration, ensuring nodes reconnect even after disruptions.

These attributes make ESP32 particularly suitable for portable, low-cost, disaster-ready communication solutions.

1.2.4. Energy and Power Management

Energy efficiency is a recurring theme in the literature, as disaster communication systems must remain operational for extended periods without reliable access to recharging infrastructure. ESP32 devices can operate for 10–15 hours in active mode and potentially months in optimized sleep configurations, depending on workload and battery capacity.

Scholarly works on energy-efficient wireless mesh networks emphasize the importance of adaptive power strategies to balance communication reliability with battery longevity, thereby ensuring sustainable operations during prolonged emergencies.

1.3 Research Gap

Although research in mesh networking and ESP32-based communication systems has advanced considerably, several critical gaps remain unaddressed in the context of disaster communication:

1.3.1. Technical Integration Deficiencies

Most existing studies isolate specific components such as ESP32 mesh networking, text-based communication, or energy optimization. However, there is a lack of end-to-end frameworks that combine portable hardware design, self-healing protocols, voice transmission and centralized monitoring into a single deployable system for real-world disaster recovery.

1.3.2. Scalability and Large-Scale Deployment

Current experiments typically involve small-scale testbeds (3 to 10 nodes). There is little empirical research exploring how ESP32 mesh networks behave when scaled to hundreds or thousands of devices across geographically distributed disaster zones. Gaps exist in analyzing routing efficiency, throughput degradation and resilience under high node density and unstable conditions.

1.3.3. Multimedia Communication Limitations

While many ESP32-based systems successfully demonstrate text messaging or sensor data transfer, compressed voice transmission and multimedia support remain underexplored. Optimizations for audio encoding, bandwidth allocation and quality-of-service (QoS) for real-time communication are largely absent in the literature, despite their critical role in disaster scenarios.

1.3.4. Configuration Management and Continuity

Studies rarely address operational continuity challenges such as automatic configuration backup, seamless node replacement and centralized configuration push. These are essential for sustaining long-duration operations where devices may fail, relocate or require reinitialization under emergency pressure.

1.3.5. Comparative Evaluation Deficiency

Research seldom benchmarks ESP32 mesh networks against alternative solutions such as satellite phones, LoRa WAN or traditional radio systems. Without systematic comparisons, the cost-effectiveness, reliability and deployment speed advantages of ESP32-based networks remain insufficiently quantified.

1.4 Research Problem

Disasters such as earthquakes, floods and severe storms frequently cause widespread failures in cellular towers, fiber backbones and power grids, leaving affected regions without reliable communication channels. This breakdown significantly disrupts rescue coordination, emergency logistics and information flow among both responders and victims. While existing alternatives such as satellite phones, professional radio systems and LoRa-based communication offer partial solutions, they suffer from several limitations including high operational costs, limited multimedia support, single-function capabilities and restricted scalability.

Although the ESP32 microcontroller and mesh networking technologies present a promising foundation for cost-effective and scalable disaster recovery communication, their application in real-world emergency contexts remains underexplored. Current research and prototypes often demonstrate only small-scale deployments, lack integration of voice and text communication and do not provide robust solutions for automatic configuration, node replacement and centralized monitoring. Furthermore, comparative evaluations with existing disaster communication solutions are scarce, making it difficult to establish ESP32 mesh networks as a viable, field-ready alternative.

This leads to the central research problem:

“How can ESP32-based self-healing mesh networks be designed and optimized to deliver reliable, scalable and infrastructure-independent communication in disaster scenarios, while overcoming the limitations of existing emergency communication systems?”

To address this overarching problem, the research is guided by the following key questions:

- **Technical Feasibility:** What design choices in terms of hardware, mesh topology and power management strategies are most effective for sustained ESP32-based disaster communication?
- **System Integration:** How can ESP32 devices combine text, voice and GPS communication with centralized monitoring and automated configuration management in a single framework?
- **Performance and Scalability:** How do ESP32 mesh networks perform under realistic disaster conditions with varying node densities, geographical spans and environmental challenges?
- **Comparative Effectiveness:** In what ways can ESP32 mesh networking outperform or complement existing disaster communication technologies in terms of cost, deployment speed, reliability and fault tolerance?

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to design, develop and evaluate a resilient ESP32-based self-healing mesh communication system that ensures reliable, scalable and cost-effective disaster recovery communication in the absence of conventional infrastructures such as cellular networks, satellites or the internet.

The system aims to provide multi-modal communication capabilities including text messaging, voice communication (broadcast and private), GPS tracking and centralized monitoring while emphasizing fault tolerance, automatic device management and remote configurability. By leveraging ESP32 microcontrollers and mesh networking protocols, the solution seeks to deliver a portable, low-cost and infrastructure-independent system for effective coordination during disaster response operations.

2.2 Sub Objectives

To achieve the main objective, the research will pursue the following sub-objectives:

1. **Develop a Self-Healing ESP32 Mesh Network**
    - Implement a distributed topology where all nodes function simultaneously as both coordinator and client.
    - Enable seamless self-organization and reconfiguration when nodes fail or disconnect, ensuring continuous communication.
2. **Implement Unique Device Identification and Tracking**
    - Assign each ESP32 device a unique identifier (via MAC address or predefined ID).
    - Enable accurate device tracking, monitoring and communication logging within the centralized dashboard.
3. **Develop a Centralized Management Dashboard**
    - Create a PC-based offline dashboard for real-time monitoring of active & inactive nodes.

- Provide visualization of network topology, communication flows and device status to improve situational awareness.

4. **Enable Remote Configuration and Automated Recovery**
   - Design mechanisms for pushing configuration updates from the dashboard to devices.
   - Implement an automatic backup & restoration system so failed nodes can be replaced seamlessly without reconfiguration.
5. **Integrate Text and Voice Communication Features**
   - Implement text-based alerts, instructions and commands sent from the dashboard to nodes.
   - Display messages on LCD screens and provide text-to-speech (TTS) playback for accessibility.
   - Support two modes of voice communication:
     - Broadcast voice (one-to-many) for emergency announcements.
     - Node-to-node voice (private) for direct field coordination.
   - Enable the dashboard to transcribe voice to text (with node-ID tagging) and replay broadcast audio.
6. **Incorporate GPS and Location-Based Features (optional improvements)**
   - Enable GPS data collection from nodes.
   - Provide map-based visualization of responder positions on the dashboard to support better coordination.
7. **Evaluate System Reliability, Scalability and Fault Tolerance**
   - Conduct performance testing under simulated disaster conditions, including node failures and harsh environments.
   - Assess scalability (up to hundreds of nodes), routing efficiency, communication range, latency and power consumption.
8. **Benchmark Against Existing Disaster Communication Solutions**
   - Compare the proposed ESP32 mesh system with satellite communication, LoRa networks and traditional radio systems.
   - Evaluation based on deployment cost, fault tolerance, speed of setup, range coverage and operational effectiveness in disaster scenarios.

3. METHODOLOGY

The methodology of this research outlines the systematic approach followed to design, develop and evaluate the proposed ESP32-based disaster recovery communication system. The methodology focuses on building a resilient, infrastructure-independent and fault-tolerant communication framework that integrates mesh networking, portable hardware and centralized management for emergency response scenarios. This section describes the system architecture, component-level design, requirements analysis, tools and technologies used, implementation strategies, deployment model and evaluation plan.

3.1. System Architecture Diagram

The overall system architecture is based on a hybrid distributed-centralized model where a network of ESP32 nodes forms a self-healing Wi-Fi mesh and a central PC dashboard acts as the command and monitoring center. Each ESP32 node operates autonomously and routes messages for others, enabling extended communication via multi-hop connections. Nodes are equipped with peripherals such as microphone, speaker, LCD display and GPS module for multimedia communication and location tracking.

Two communication modes are supported: (1) voice broadcast, where a node's audio is transmitted to all other nodes and the dashboard and (2) node-to-node private voice communication. The central dashboard visualizes network topology, monitors node activity, logs message and converts voice data to text tagged with sender IDs. The mesh routing protocol provides inherent fault tolerance by dynamically rerouting traffic when nodes fail.

Diagram Description: The architecture diagram illustrates multiple ESP32 nodes interconnected in a mesh network with bi-directional multi-hop links. The dashboard is connected to the mesh via a coordinator node, providing centralized visualization and control. Data flows (voice, text, GPS and config) are represented along these paths, highlighting the redundancy and fault tolerance through multiple routes.

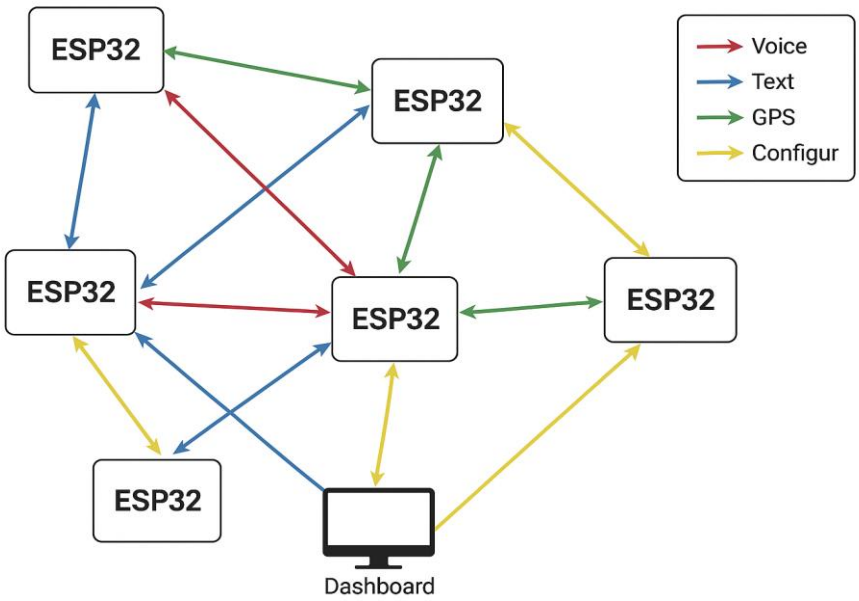

Figure 1: System Diagram

3.2. Component-Specific System Diagram

At the hardware level, each node integrates several components centered on the ESP32 microcontroller:

- **ESP32 Microcontroller:** Handles Wi-Fi mesh communication, processes routing and message protocols and manages connected peripherals.
- **LCD Display:** Shows text messages, alerts and node status to field users.
- **Text-to-Speech (TTS) Converter:** Converts text messages into audio for playback via the speaker.
- **Microphone & Speaker:** Enable bidirectional voice input/output; audio is digitized, compressed, transmitted and played locally.
- **GPS Module:** Provides real-time location data sent to the dashboard for mapping.

- **Battery and Power Management:** Rechargeable lithium-ion battery supplies power with charging circuits and low-power operational modes for extended use.
- **Flash Memory or SD Storage:** Used for storing configuration backups, communication logs and temporary data for node recovery.

Diagram Description: The component diagram shows the ESP32 at the center, linked to the Central dashboard, LCD display, microphone, speaker, GPS module, battery and storage modules, detailing the functional interconnections.

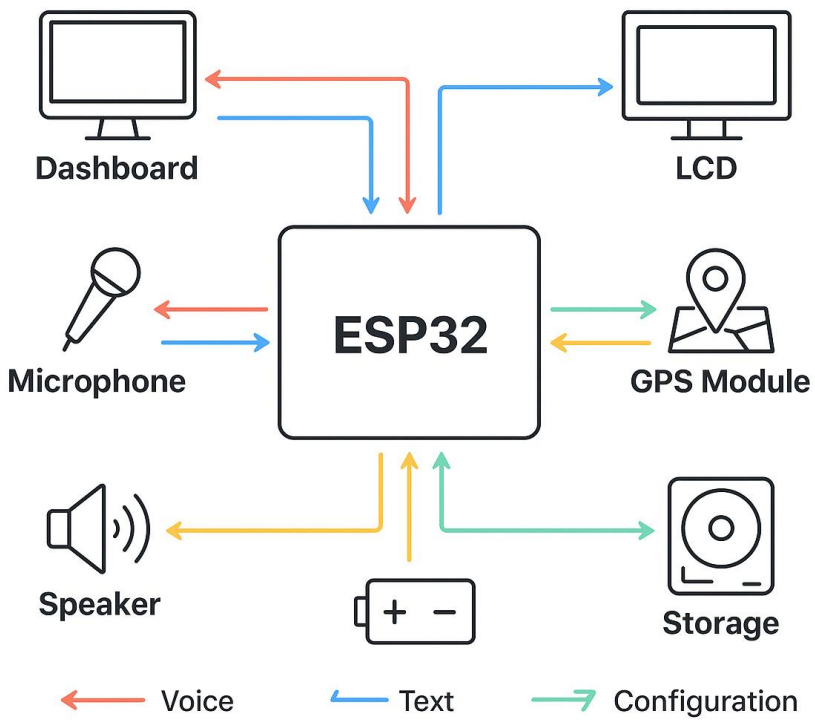

Figure 2: Component-Specific System Diagram

3.3. Project Requirements

3.3.1. Functional Requirements

The system is designed to fulfill the following functional requirements:

1. Automatic formation of a self-healing Wi-Fi mesh network between nodes.

2. Bidirectional text messaging with dashboard-to-node alerts displayed on LCDs and optionally read aloud.
3. Support for voice broadcast and private node-to-node voice communication.
4. Voice-to-text conversion with messages tagged by device ID for dashboard display.
5. GPS tracking data transmitted to the dashboard for situational awareness.
6. Unique device identification per node using MAC address or preset IDs.
7. Centralized dashboard providing real-time monitoring, fault alerts and configuration management.
8. Automated backup and restore functionality for node configurations enabling seamless node replacement.

3.3.2. Non-Functional Requirements

1. **Scalability** – System must support hundreds of nodes while maintaining efficiency.
2. **Fault Tolerance** – Communication must persist despite multiple node failures.
3. **Security** – Implement lightweight encryption to secure sensitive communications.
4. **Usability** – Simple interface for responders with both visual (LCD) and auditory (TTS/voice) outputs.
5. **Offline Operability** – Dashboard should work without internet connectivity.

3.4. Tools and Technologies

3.4.1. Tools (Free and Open-Source)

1. **Arduino IDE:** For firmware writing and debugging on ESP32 devices.
2. **VS Code:** Advanced code editing and debugging environments.
3. **Wireshark(optional):** For monitoring network traffic and debugging mesh communication.
4. **Node-RED or Python Visualization:** For prototype dashboard development and real-time visualization.

5. **QEMU-ESP32 Emulator:** To simulate firmware for early testing before hardware deployment.

3.4.2. Technologies

- **ESP-WIFI-MESH:** Native ESP32 protocol for self-healing wireless mesh networking.
- **SQLite or Local Database:** For local storage of logs and messages on the dashboard system.
- **Python (Tkinter or PyQt):** GUI frameworks for building the dashboard interface.
- **TTS & STT Libraries:** eSpeak for text-to-speech and Vosk for offline speech-to-text conversion.
- **MQTT (optional):** Lightweight messaging protocol for structured data exchange between components.
- **Compression Algorithms:** To optimize voice data transmission bandwidth.

3.5. Implementation

The project implementation is conducted in structured phased development cycles, ensuring robust development, integration and refinement of each system component:

- **Node Firmware Development:** The core phase involves developing custom firmware for ESP32 nodes to support ESP-WIFI-MESH based self-healing mesh networking functionalities. This includes implementing message routing protocols, seamless GPS module integration and automatic backup and restore systems to preserve device configurations. The firmware also manages voice and text communication routing.
- **Peripheral Integration:** Hardware peripherals are connected and calibrated to ensure seamless functionality. The LCD screens are integrated for on-node display of incoming alerts and messages; microphones and speakers are connected for voice input and output, enabling both broadcast and private voice communication. The GPS modules are calibrated for accuracy in location tracking and power management circuits are set to optimize battery usage.

- **Dashboard Development:** A graphical user interface (GUI) application is developed using free tools like Node-RED or Python GUI frameworks (Tkinter or PyQt). This dashboard provides real-time visualization of the mesh network topology, live message and voice management, node health monitoring and configuration controls. It also includes voice-to-text conversion to transcribe audio communications for situational awareness.
- **Voice/Text Processing Integration:** Offline Text-to-Speech (TTS) and Speech-to-Text (STT) engines, such as eSpeak and Vosk, are integrated into the system to enable real-time voice and text interaction between responders and the command center.
- **Prototype Testing:** An initial small-scale testing phase is conducted with 10–20 ESP32 nodes. This phase validates basic functionalities, communication reliability, hardware integration and dashboard operations under controlled conditions.
- **Scalability Trials:** The network scale is increased progressively beyond 100 nodes to evaluate performance under higher loads. This phase assesses throughput, latency, fault tolerance and mesh self-healing efficiency to ensure the system remains effective at real-world operational scales.
- **Power Management Enhancement:** While hardware-based power management mechanisms are already implemented on the nodes, this project aims to improve these by enabling centralized power mode management through the command dashboard. Each node runs on rechargeable lithium-ion batteries and through the dashboard, power states such as active, modem sleep, light sleep and deep sleep can be remotely configured to optimize battery life and operation according to field needs.

3.6. Deployment

The deployment strategy is designed to enable rapid and effective field readiness during disaster recovery operations:

- **Field Node Distribution:** Portable ESP32 mesh nodes are distributed strategically across disaster-impacted areas by emergency responders and volunteers. Upon powering on, these nodes autonomously form an ad hoc, self-healing wireless mesh network providing resilient communication coverage over the affected geography.
- **Central Command Setup:** One ESP32 node is designated as the coordinator or root node and is positioned near the command center. This node acts as a gateway linking the mesh network to a centralized monitoring dashboard running on a ruggedized laptop or PC capable of functioning offline.
- **Power Supply:** Each node operates on rechargeable lithium-ion batteries, ensuring the system remains independent of potentially disrupted grid power. Optional solar charging units may be incorporated for extended field operations enabling sustained uptime.
- **Network Expansion:** Additional nodes may be introduced dynamically during deployment to naturally extend mesh coverage and reinforce network robustness without complex configuration requirements.
- **User Interaction:** Emergency response teams, volunteers and affected residents use the deployed nodes for voice and text communication, GPS tracking and receiving alerts, facilitating coordinated rescue and relief efforts even in the absence of conventional communication infrastructure.

3.7. Evaluation of the System

A comprehensive multi-dimensional evaluation approach will be undertaken to validate the system's effectiveness and operational readiness:

- **Reliability Testing:** Systematic measurement of message delivery success rates and end-to-end latency for text and voice communications across different node densities to verify consistent network operation.
- **Scalability Analysis:** Evaluation of performance metrics including throughput, latency, routing efficiency and network stabilization as the node count increases, ensuring the mesh network maintains robustness under stress.
- **Fault Tolerance Assessment:** Controlled simulation of node failures and communication interruptions to validate the self-healing capability of the mesh protocol—assessing how rapidly and effectively the network reroutes traffic to maintain continuous connectivity.
- **Voice Quality Testing:** Assessment of broadcast and node-to-node voice communication for clarity, transmission latency and intelligibility, ensuring that voice messages remain understandable and timely.
- **Energy Efficiency Measurement:** Monitoring of battery consumption across diverse operational scenarios and power modes, with particular attention to the effectiveness of remotely managed power-saving configurations to prolong node uptime.
- **Usability Testing:** Gathering qualitative and quantitative feedback from dashboard operators and field users relating to the system's ease of use, interface intuitiveness and physical device ergonomics to identify areas for improvement.
- **Comparative Benchmarking:** Benchmarking the proposed ESP32-based system against existing emergency communication technologies such as satellite phones, LoRa mesh networks and traditional radio systems. The comparison includes critical factors like per-node cost, deployment speed, fault tolerance, communication range and operational reliability to position the solution in the wider emergency response ecosystem.

This detailed evaluation protocol underpins the project's holistic engineering approach—balancing rigorous functional validation with practical usability and field deployment readiness, thereby ensuring the ESP32 disaster recovery mesh communication system meets the demanding requirements of real-world emergency operations.

4. WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure (WBS) for this ESP32-based disaster recovery communication system project outlines the major tasks and sub-tasks involved from initiation to final validation, ensuring a structured approach for systematic development and delivery.

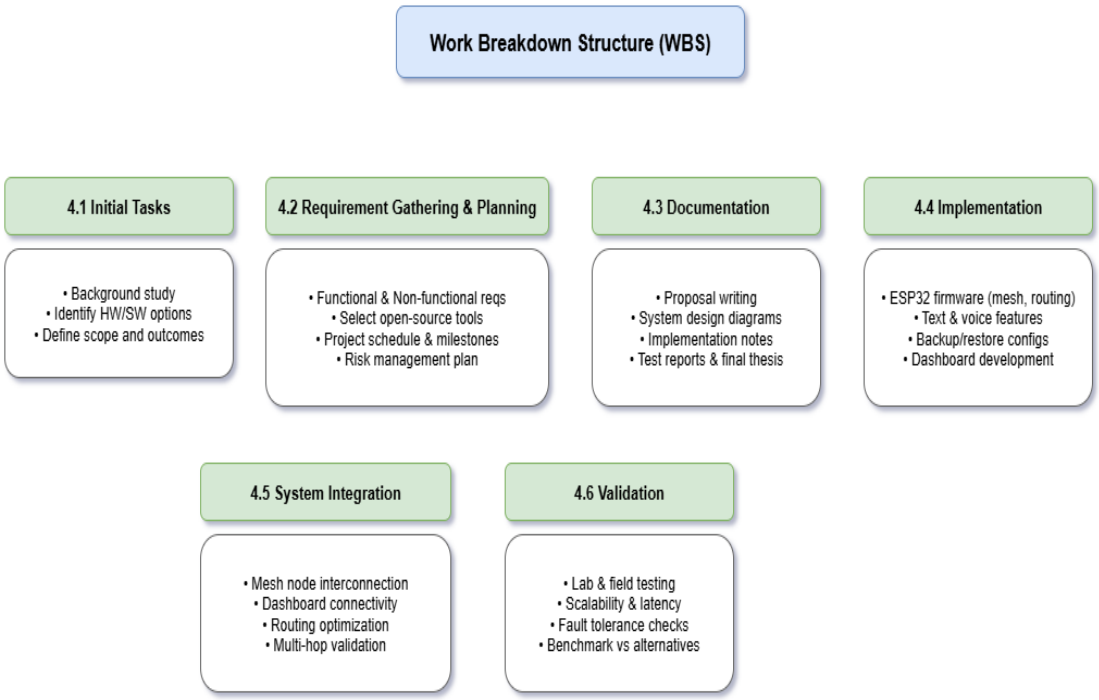

Figure 3: Work Breakdown Structure

4.1. Initial Tasks

- Conduct preliminary background study on disaster recovery communication technologies.
- Identify potential hardware platforms (ESP32, sensors, displays, GPS modules, etc.).
- Review open-source software tools and libraries suitable for mesh networking, voice and text processing.
- Establish project scope, limitations and expected outcomes.
- Formulate research questions and refine objectives.

4.2. Requirement Gathering and Planning

- Collect technical requirements for ESP32 mesh communication, text and voice features.
- Define functional requirements (messaging, voice broadcast, node-to-node calls, GPS tracking, dashboard logging).
- Define non-functional requirements (scalability, reliability, cost, power efficiency, usability).
- Identify free/open-source development tools and frameworks.
- Develop project schedule and allocate milestones for each phase.
- Design risk management strategy (e.g., hardware failures, time delays, integration issues).

4.3. Documentation

- Prepare Project Proposal Document including problem statement, literature review, methodology and objectives.
- Maintain Design Documentation: system architecture diagrams, node-level component design and data flow specifications.
- Draft Implementation Documentation covering firmware logic, dashboard architecture and communication protocols.
- Maintain Testing & Validation Reports to record evaluation results.
- Prepare Final Research Thesis Report consolidating all findings, evaluations and contributions.

4.4. Implementation

- Develop ESP32 firmware for mesh networking (self-healing, routing, message handling).
- Integrate peripheral devices: microphone, speaker, LCD, GPS and storage.
- Implement text messaging with LCD display + text-to-speech output.
- Implement voice communication system (broadcast + node-to-node).
- Develop voice-to-text processing and message logging on dashboard.
- Implement backup/restore functionality for node configurations.

- Develop centralized management dashboard (Python GUI with SQLite database).

4.5. System Integration

- Connect multiple ESP32 nodes into a working mesh environment.
- Link nodes with the dashboard for real-time monitoring.
- Test device identification, tracking and GPS mapping functions.
- Validate bidirectional communication flows (text-to-voice, voice-to-text, dashboard-to-node, node-to-dashboard).
- Optimize routing and fault tolerance in multi-hop communication scenarios.
- Ensure cross-compatibility of dashboard with different OS platforms (Windows/Linux).

4.6. Validation

- Conduct controlled lab tests with small-scale networks (5–10 nodes).
- Perform scalability tests with larger simulated networks (50–100+ nodes).
- Measure system performance (latency, packet delivery ratio, throughput, power consumption).
- Test fault tolerance by simulating node failures and verifying automatic rerouting.
- Evaluate usability by collecting feedback from simulated responders using the system.
- Benchmark against existing disaster communication methods (satellite, LoRa, radio).
- Document validation outcomes and refine system based on test results.

This breakdown provides a clear roadmap from initial research and planning through iterative development, integration and rigorous validation for delivering a reliable ESP32 disaster recovery mesh communication system.

5. PROJECT DELIVERY PLAN

The following Gantt charts show the project schedule we are going to work through, displaying the major milestones of the entire project activities and time span of each of these activities respectively. Such visual representation will aid in monitoring activities, expense management as well as the realization of every stage of the project on time.

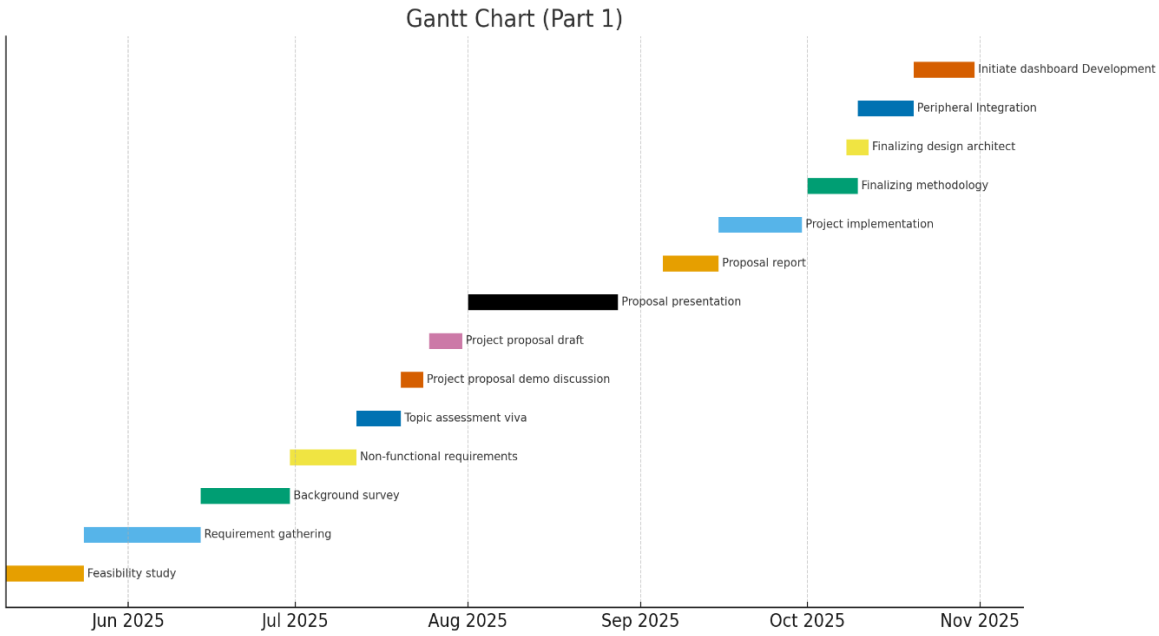

Figure 4: Grantt chart part 01

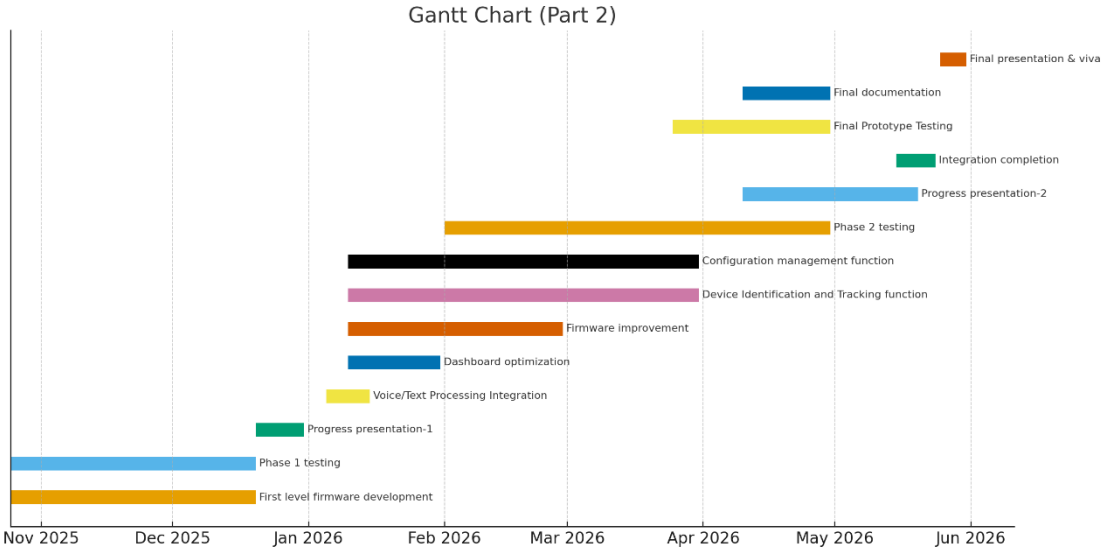

Figure 5: Grantt chart part 02

6. COMMERCIALIZATION OF THE PRODUCT

The commercialization of the ESP32-based disaster recovery communication system focuses on transforming the developed technology into a viable product that meets the urgent needs of emergency response organizations, government agencies, non-governmental organizations (NGOs) and communities in disaster-prone regions worldwide. The product's inherent advantages of affordability, portability, infrastructure, independence and resilience position it strongly within the growing market for emergency communication solutions.

6.1. Target Market

The primary target market for this product includes:

- **Emergency Response Agencies:** Fire departments, disaster management authorities, search and rescue teams and first responders who require reliable communication in disaster scenarios where traditional communication infrastructure fails.
- **Government and Municipal Disaster Management Units:** Local, regional and national government bodies tasked with disaster preparedness and recovery operations, especially in regions prone to earthquakes, floods, hurricanes and other natural calamities.
- **Non-Governmental Organizations (NGOs) and Humanitarian Aid Groups:** Organizations delivering rapid disaster relief and requiring coordination tools that function without relying on cellular or satellite networks.
- **Rural and Remote Communities:** Regions with limited traditional communication infrastructure, were infrastructural outages during disasters severely hinder emergency communication.
- **Private Sector Infrastructure Companies:** Companies managing large-scale critical infrastructure (e.g., power grids, transport networks) that require contingency communication mechanisms during emergencies.

6.2. Revenue Model

The revenue generation strategy for the ESP32 disaster recovery communication system includes:

1. **Direct Product Sales:** Selling portable ESP32 communication units and coordinator kits to emergency response agencies, NGOs and governmental organizations either as one-time purchases or through contracts.
2. **Software Licensing and Support:** Offering licensed software packages for the command center dashboard with options for custom features, updates and technical support subscriptions.
3. **Training and Setup Services:** Providing professional training, system installation and integration services to ensure customers can rapidly deploy and operate the system effectively.
4. **Custom Development and Consulting:** Tailoring the solution to specific organizational needs, including integration with existing emergency communication infrastructure and consulting on disaster preparedness communication strategies.
5. **Partnerships and Grants:** Collaborating with governmental bodies and international aid organizations to secure funding, grants, or subsidized sales aiming at widespread community adoption in vulnerable areas.

6.3. Implementation Plan

To successfully commercialize the product, a phased implementation plan is proposed:

Phase 1	Pilot Deployments and Validation	Partner with select emergency response organizations and NGOs to deploy pilot systems in real-world disaster exercises or controlled environments. Collect user feedback and data to refine hardware, software and usability aspects.
Phase 2	Production Scaling and Quality Assurance	Production Scaling and Quality Assurance: Establish manufacturing partnerships for mass production of ESP32 nodes and peripherals with quality control measures. Finalize software packaging and documentation
Phase 3	Market Launch and Outreach	Initiate targeted marketing campaigns, participate in disaster preparedness expos and conferences and build relationships with key industry stakeholders. Leverage case studies from pilot deployments demonstrate effectiveness.
Phase 4	Training and Customer Support Program	Develop comprehensive training modules and offer ongoing support, both remotely and on-site, to maximize user adoption and satisfaction.
Phase 5	Continuous Improvement and Feature Expansion	Use feedback from early adopters for iterative product enhancement, including adding advanced features like AI-powered network optimization and multi-protocol communication compatibility.

Table 1: Implementation Plan

By addressing the specific needs and pain points in disaster communication with a cost-effective, easy-to-deploy and robust system, this commercialization plan aims to ensure widespread adoption of the ESP32 mesh communication system, improving emergency response outcomes globally.

7. BUDGET AND BUDGET JUSTIFICATION

The budget for this research project is structured to support the successful completion of all phases, including requirement analysis, prototype development, system testing, documentation and dissemination. Emphasis is placed on maintaining cost-effectiveness while ensuring that all necessary resources are available to demonstrate the feasibility and practicality of the proposed disaster communication system.

7.1. Estimated Budget Breakdown

Category	Total (LKR)
Software & Tools	10,000
Hardware & Accessories	30,000
Testing & Evaluation	10,000
Documentation & Reports	5,000
Project Management & Miscellaneous	5,000
Total	**60,000**

Table 2: Budget allocation chart

7.2. Budget Justification

The budget is designed to ensure a balanced allocation of resources across the project lifecycle rather than heavily focusing on hardware alone.

- **Software & Tools (10,000 LKR):** While most development tools (Arduino IDE, Python, Node-RED) are open-source, a portion of the budget is allocated for paid simulation environments, plug-ins and data visualization tools required to design and evaluate the system effectively.
- **Hardware & Accessories (30,000 LKR):** A minimal hardware setup (ESP32 boards, basic sensors, batteries, LCD displays) is included only to demonstrate system functionality. The focus is on validating communication protocols and dashboard monitoring, not on building large-scale hardware systems.
- **Testing & Evaluation (10,000 LKR):** Essential for validating performance in simulated disaster environments. Covers costs for controlled test setups, travel for

outdoor range testing and collection of performance data (latency, throughput, fault tolerance).

- **Documentation & Reports (5,000 LKR):** Professional printing, binding and report preparation are crucial for academic submission and dissemination of research outcomes.
- **Project Management & Miscellaneous (5,000 LKR):** Allocated to handling unforeseen expenses such as replacement parts or backup components. It also covers small-scale logistical needs during project execution.

This distribution ensures that the budget is research-driven and emphasizes system design, software development and evaluation while keeping hardware costs minimal. The approach supports creation of a functional, validated prototype with practical results, while maintaining affordability and academic feasibility.

8. REFERENCES

1) *D. Sutradhar, "LoRa-Powered IoT Messaging System for Internet-Free Communication," International Journal of Advanced Research in Science, Engineering and Technology, 2024.*
2) *R. Manoj and D. Baker, "Communication Challenges in Emergency Response," Communications of the ACM, vol. 50, no. 3, pp. 51-56, 2007.*
3) *P. Nguyen et al., "MeshSOS: An IoT Based Emergency Response System," University of Hawaii ScholarSpace, 2020.*
4) *Espressif Systems, "ESP-WIFI-MESH Programming Guide," Official Documentation, 2024.*
5) *S. Eaton et al., "Emergency Network Deployment: Mesh in Disaster Management," Meshmerize Research, 2024.*
6) *C. Schmidt et al., "BPoL: A Disruption-Tolerant LoRa Network for Disaster Response," PeaceSec Journal, 2024.*
7) *Sabari et al., "LoRaLink: Resilient Communication for Firefighters via Meshtastic Mesh Networks," University of Twente, 2024.*
8) *K. Bharti, "Performance Evaluation of ESP32 in Outdoor Links," Brazilian Telecommunications Journal, 2024.*
9) *H. Zha et al., "Solutions for Sustainable and Resilient Communication in Disaster Management," arXiv preprint, 2024.*
10) *J. Manoj and D. Baker, "Case Studies of Communication Systems During Harsh Environments," Safety Science, 2019.*